



S.1 TEAM INTRODUCTION & TEAM STRUCTURE

Team Structure

Subsystem	Head	Associate
Design	Aniket Sapkale	Sayali Zolekar
Steering	Parth Chennur	Mandeepsingh Loniwale, Vivek Jagdale
Suspension	Parth Chennur	Mandeepsingh Loniwale, Vivek Jagdale
CAE	Adwait Lawate	Khushi Patil
Braking	Omkar Kale	Sarthak Phantagare
Electrical	Mugdha Ganu	Aryan Khalane
Autonomous	Jayesh Sonawane	Aditya Raut, Durvesh Vikhankar
Solar	Aditi Panganti	Aryan Khalane
Manufacturing	Sankalp Wasnik	Aniket Sapkale, Omkar Kale, Mandeepsingh Loniwale
Cost	Adwait Lawate	Sarthak Phatangare
Business & Media	Aditi Panganti	Omkar Kale
HVAC	Adwait Lawate	Vivek Jagdale

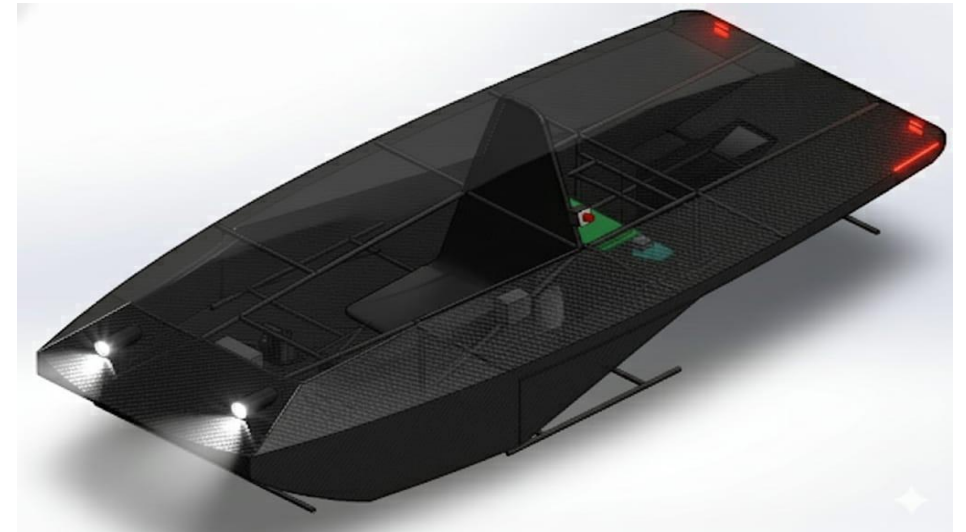
Team Name: STES Hyperion

College Name: Sinhgad College Of Engineering, Vadgoan BK.

Vehicle Category: Hyper Electric Vehicle (HEV)



**WE CLUTCH !
WE ENDEAVOUR !
WE ARE HYPERION !**





S.2 PROJECT PLAN

Stage 1: Team Formation (MAY-JUN 2025)

- 1) Recruited new members for different subsystems.
- 2) Conducted basic training and assigned tasks based on the previous rule book.

Stage 2: Registration and Sponsor hunt (JULY 2025)

- 1) Our team has been successfully registered for SEVC 2026 in the HEV category, marking the start of our journey.

S 3: Design Phase (AUG-SEPT 2025)

- 1) With the updated rulebook in our hand, we focused on making the vehicle more compact & lighter.
- 2) With these goals chassis, suspension & braking were designed.

Stage 4: Manufacturing Phase (OCT-JAN 2026)

- 1) Built a mock chassis, identified areas for improvement, and revised designs.
- 2) Currently in the manufacturing stage, with testing planned next.

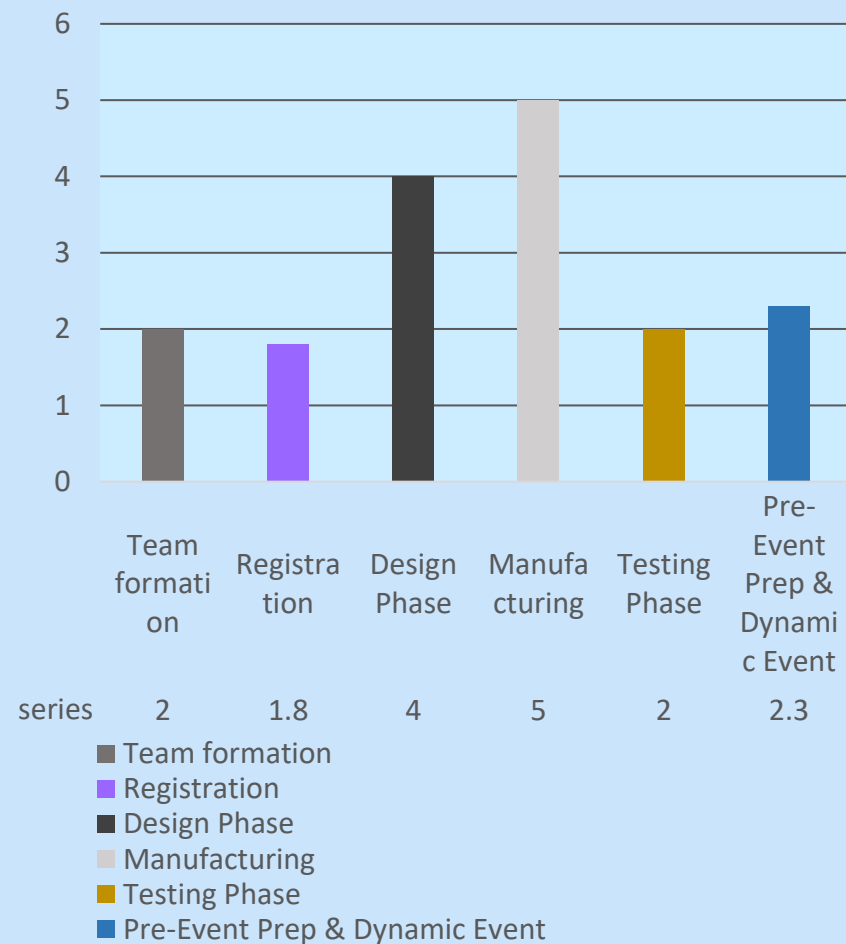
Stage 5: Testing phase (JAN 2026)

- 1) We conducted vehicle testing to validate performance, safety, and compliance, with SEVC standards

Stage 6: Pre-Event Prep & Dynamic Event (FEB – MAR 2026)

- 1) After complete testing, the vehicle will be shipped to the event location.
- 2) We are gearing up to deliver our best during 5-day SEVC Dynamic Events.

Duration (Months)





S3. 3D Assembly Vehicle View

3 A – 3D Assembly Vehicle View

Use of primary and secondary member classification for weight optimization

Designed considering manufacturability and ease of assembly

Adequate factor of safety maintained under expected loading conditions

Design compliant with SEVC structural and safety regulations

- ROLL CAGE Dimensions : Height-1100mm, Width- 650mm, Length- 2185.7mm
- Wheel Base : 2160mm
- Track Width : 1300mm

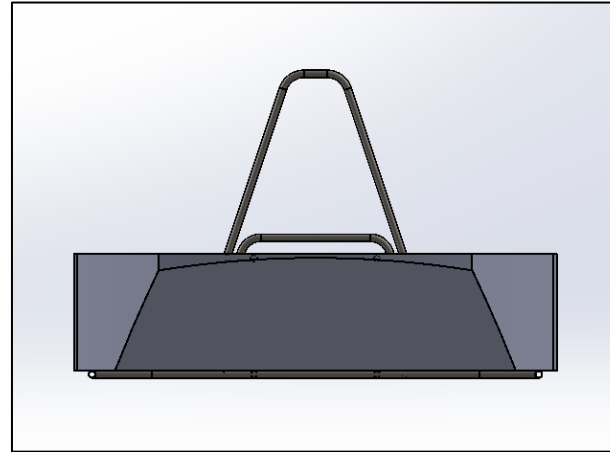


Fig 3.1 Front View

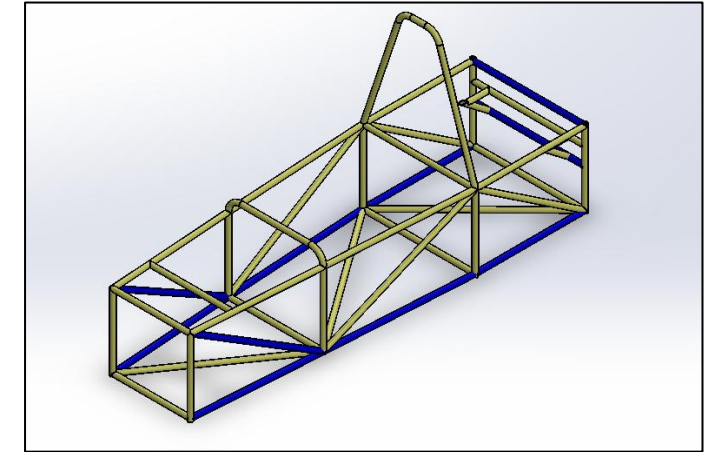


Fig 3.2 Isometric view

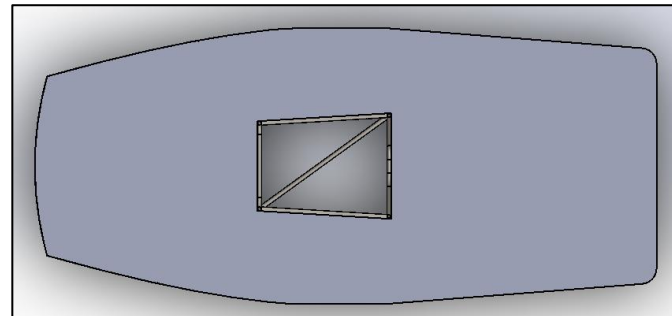


Fig 3.3 Top View

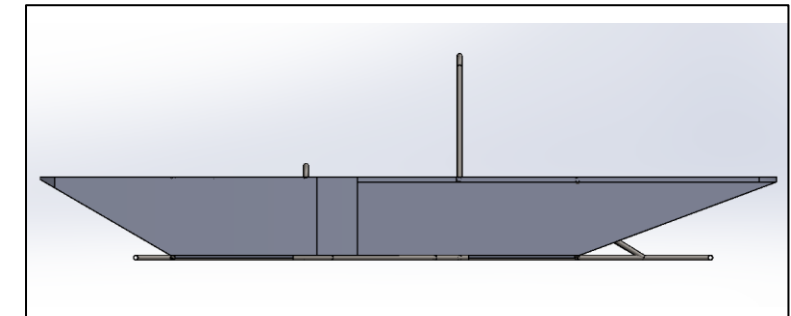


Fig 3.4 Side View



S.3 Roll Cage Iterations

3 B – Roll cage Design Iterations

**Material Used :- AISI 4130
(Chromoly Steel)**

Technical Property	Value	
Thickness	For 1.2 mm members	For 1.6 mm members
1.Bending Strength	737.27 Mpa	760 Mpa
2. Stiffness	1.40×10^9	1.78×10^9
3.Yield Strength	737.27 Mpa	760Mpa

Iteration 1

● **Challenge:** High bending stress and low stiffness in initial roll cage layout.

✓ **Solution:** Identified critical load-bearing members for optimization.

Iteration 2

● **Challenge:** Improving stiffness without major weight increase.

✓ **Solution:** Increased thickness of primary members to 1.6 mm and improved load paths.

Iteration 3

● **Challenge:** Excess weight due to uniform thick sections.

✓ **Solution:** Optimized design using 1.6 mm (primary) and 1.2 mm (secondary) members meeting SEVC safety requirements.



S.4 CAE Analysis of Vehicle/Frame

• COMPUTER AIDED ENGINEERING (CAE)

- 1) Virtual simulation and evaluation of vehicle/ frame performance.
- 2) Reduction in development time, cost, and design errors

• EXPLICIT DYNAMIC

- 1) Accurate simulation of crash, impact, and high-speed dynamic events.
- 2) on of large deformation and failure behaviour.

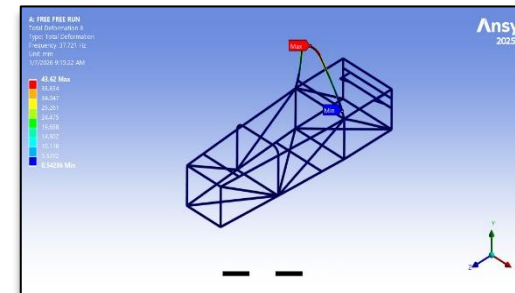
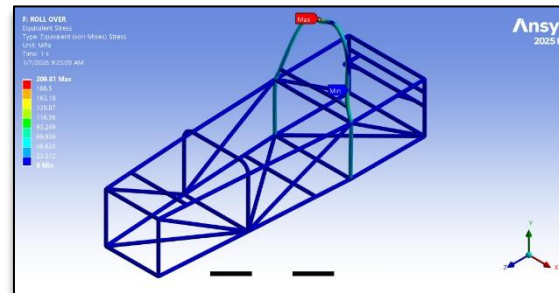
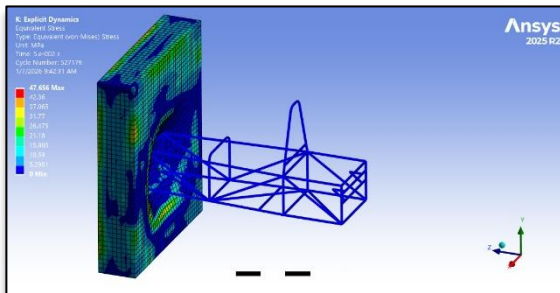
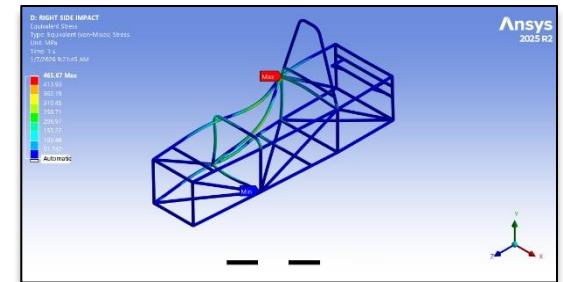
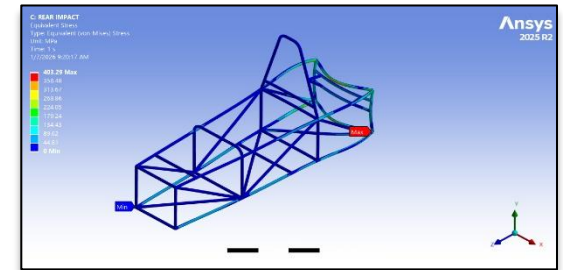
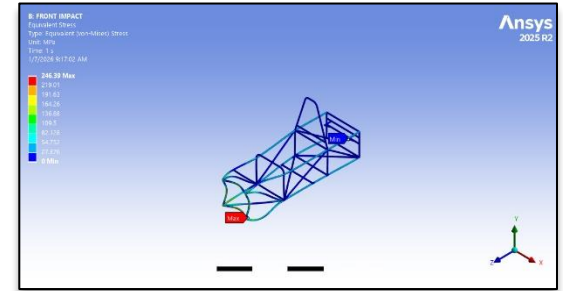
• STATIC ANALYSIS

- 1) Assessment of stresses and deformation under constant loading.
- 2) Verification of structural strength and safety

• MODAL ANALYSIS

- 1) Identification of natural frequencies and mode shapes.
- 2) Prevention of resonance and improvement of vibration performance

TYPE OF IMPACT	BOUNDRY CONDITION	TOTAL DEFORMATI ON (mm)	FOS	STRESS (Mpa)
FRONT IMPACT	Rear fixed – Front impact	2.2169	2.99	246.39
REAR IMPACT	Front fixed – Rear impact	5.5603	1.82	403.29
SIDE IMPACT	Lateral fixed – Opposite impact	10.06	1.58	465.67
ROLLOVER	Bottom fixed – rollover impact	1.312	3.51	209.81



Yield Strength

737.24 MPa

Ultimate Yield Strength

783.39MPa



S.5 SUSPENSION SYSTEM

Types of Suspension

Front

- Double Wishbone



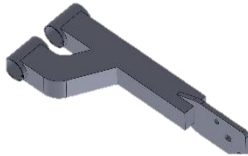
Front suspension wishbone



Front Suspension knuckle

Rear

- Trailing Arm



Trailing arm

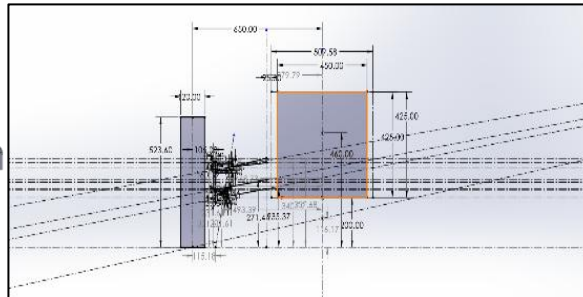
Rear ride rate :25.03N/mm

Front Ride Rate :20.68N/mm

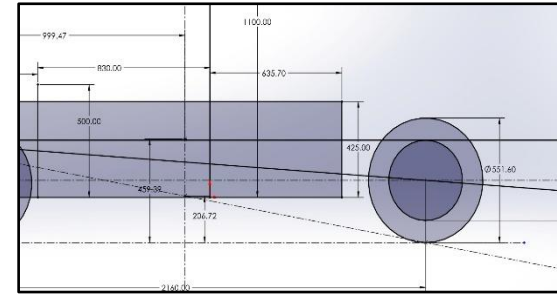
CG Height:- 459.39mm

Ground Clearance:- 200mm

Wheel Travel:- 120mm



Front View Geometry



Rear Geometry

Spring Specification

Parameters	Front	Rear
Material	Cold drawn spring steel G=81370N/mm	Cold drawn spring steel G=81370N/mm
Stiffness	40.2N/mm	49.04N/mm
Wire Diameter	7mm	7mm
Mean Diameter	48mm	48mm
No. of Coil	4	4

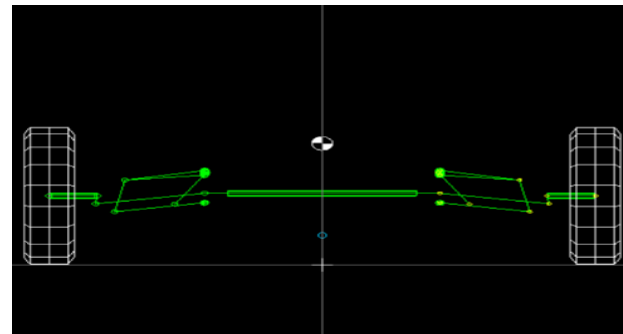
Anti- Squat	45%
Anti-Dive	37.30%
FVSA length	8563.47mm

Sprung mass

224.6 kg

Un-sprung mass

58.4 kg



Front Suspension Simulation

Bump condition changes

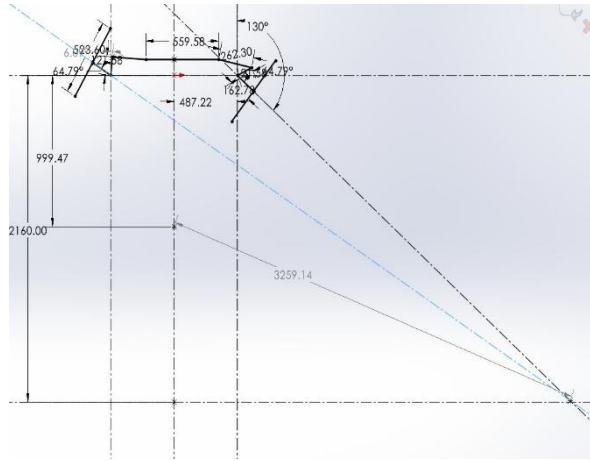
Parameters	
Caster	6.6
Camber	0
Toe	0

Parameters	
Camber	-0.63 to 0.26
Toe	-0.61 to 0.14



S.6 STEERING SYSTEM

Steering Geometry :



Wheelbase :2160mm

Track width:1300mm

Caster:6.6

KPI:11.6

Scrub Radius:115.18mm

Ackermann Percentage :99.53%

Steering wheel Di. :275mm

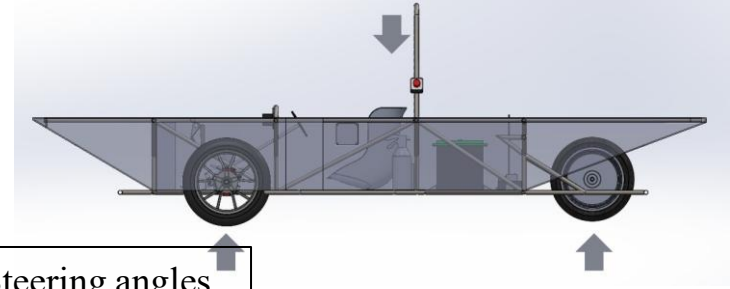
Turning Radius :3.25m

Steering Effort :36.82N

Steering ratio = 4.3

Rack travel:67.16 mm

Steering Effort Calculation :



Steering angles

Inside angle

40°

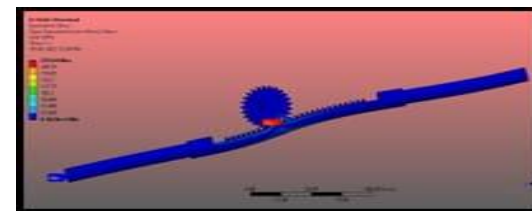
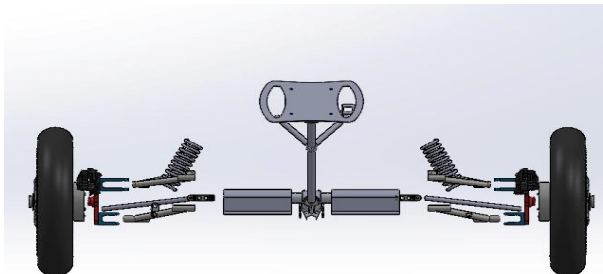
Outside angle

31.58°



Casing

	Front	Rear
Parameters	Specification	Specification
No. of wheels	2	1
Wheel Type	Alloy Rim	Hub motor
Rim Dia	14 Inch	14 Inch
Tyre Type	Tubeless	Tubeless
Tyre Size	120/70 R14	140/70 R14
Load Rating	218kg	265Kg
Inflation Pressure	28-33 psi	30-35 psi



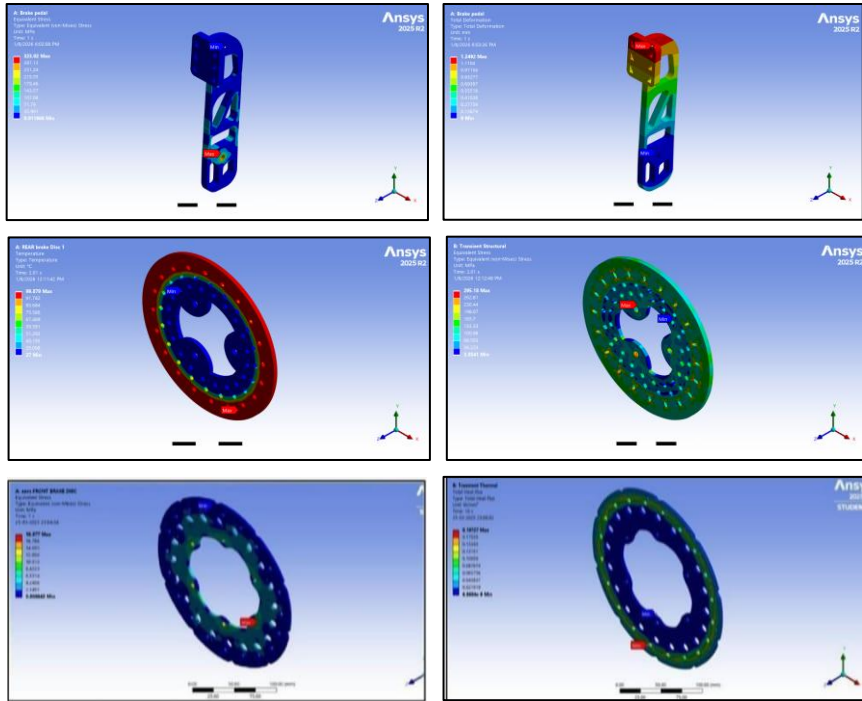
Rack and Pinion Analysis

Component and Material

Rack and Pinion	EN24
Steering Casing	AL 2024
Tie Rod	AL 2024
Steering column	AL 2024
Steering arm	MS
Steering Wheel	Carbon Fiber



S.7 BRAKING SYSTEM

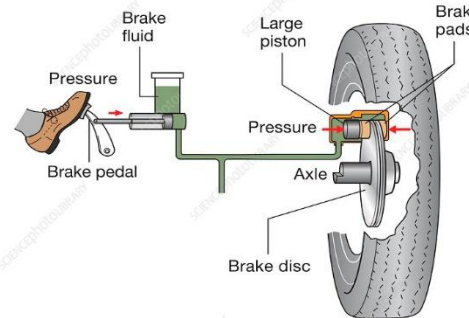


Brake Disc Diameter :
(Material : SS 410)

Front : 220mm

Rear : 200mm

Schematic diagram of braking



Stopping Distance : 7m

➤ Important Calculations :

Force applied on pedal : 1000N

Clamping Force : 2736.84 N (front)

Front and Rear Static load Split : 54% & 46%

Front and Rear Brake Force : 69% & 31%

Braking torque Generated : 268.21 Nm (front)

Braking torque Generated : 481.68 Nm (Rear)

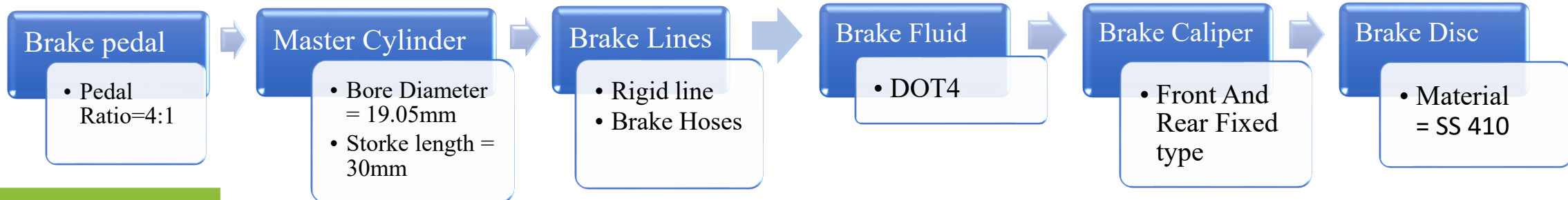
Coefficient of Friction Between Road & Tyre : 0.75

Coefficient of Friction Between Pad & Disc : 0.45

Brake Pedal Height & Travel : 300mm & 90mm

Area of Brake Pad: 1250mm²

➤ Braking System Works By :





S.8.1 Powertrain Calculations & Electrical

➤ Considerations for Transmission unit:-

- **Starting Torque:-** 60.37 Nm

➤ Transmission unit :-

- Type of Transmission : Hub
- **Tyre size:** Front : 523.6 mm
Rear : 551.6mm
- **Max Power on wheel :-** 3910W

➤ Performance

- Acceleration : 0.7m/s^2
- Maximum Speed of the Vehicle :- 63.83 km/hr

NVH CONSIDERATION:

Wiring: Cable ties and heat-shrink sleeves.
Battery : Rubber and foam isolation

➤ Motor Specifications :-

Type : Hub Motor
Voltage : 48V
Rated Power : 3000W
Maximum Power : 4000W
Rated Torque : 44 Nm
Maximum Torque : 184Nm
Continuous Current : 50 A
Max Current : 75A
Rated Speed : 450RPM
Maximum Speed : 900RPM
Insulation Class : Class B (Up to 130°C)

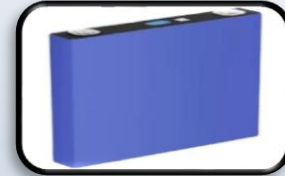


➤ Tractive Accumulator specifications :-

- Battery Type : Lithium Ferro Phosphate
- Battery Voltage : 51.2 V
- Battery capacity : 100 Ah
- C-rate :- 3C
- Cell Configuration :- 16S1P
- Number of Battery pack :- 01

➤ Cell specification :-

- Cell chemistry :- LiFePO4
- Type :- Prismatic cells
- Voltage : 3.2V
- Cell Capacity:- 100 Ah



➤ DC to DC Converter :-

- Type of Converter : Buck Converter
- Type :- Isolated
- Input Voltage : 24V-72V
- Output Voltage : 12V
- Current Rating : 10Ah
- Efficiency : 92%



➤ GLV Battery specifications :-

- Battery Type : Lead Acid
- Battery Voltage : 12 V
- Battery capacity : 10 Ah

Parameter	value
Voltage Range	48V - 72V
Continuous Current	90 A
Peak Current	270 A
Power	4320 Watt
Semiconductor	MOSFET
Control Type	Sinusoidal Control

Motor Controller :

KLSN7230N Sinusoidal
Motor Controller



S.8.2 Powertrain Calculations & Electrical

➤ Charger specifications :-

- Input voltage :- 230V AC
- Output Voltage :- 58.4V
- Current Rating :- 20A



SSR

Input	3 – 32V DC
Output	4 - 60V DC
Current Rating	60 DD

➤ SOC Calculation Method :-

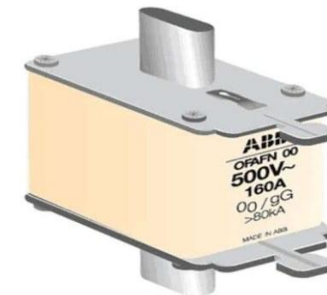
State of Charge (SOC) represents the available capacity of the battery expressed as a percentage of its nominal capacity. We used Coulomb Counting method to estimate the SOC.

$$SOC = SOC_{initial} - (I * t / c) * 100$$



DC MCCB

Voltage rating	220 V DC
Current Rating	160 A
Ultimate breaking capacity	10kA

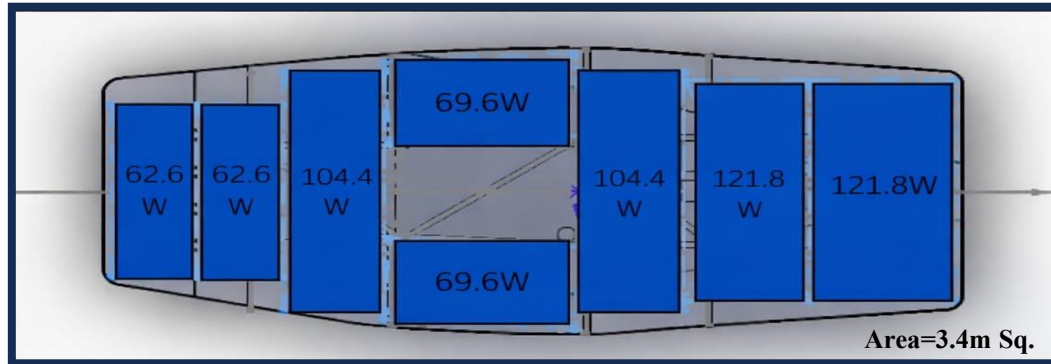


HRC Fuse

Voltage Rating	500V AC
Current Rating	160A
Rapturing Capacity	8KA



S.9 Solar Power System , Body Panels



➤ Solar Panel :

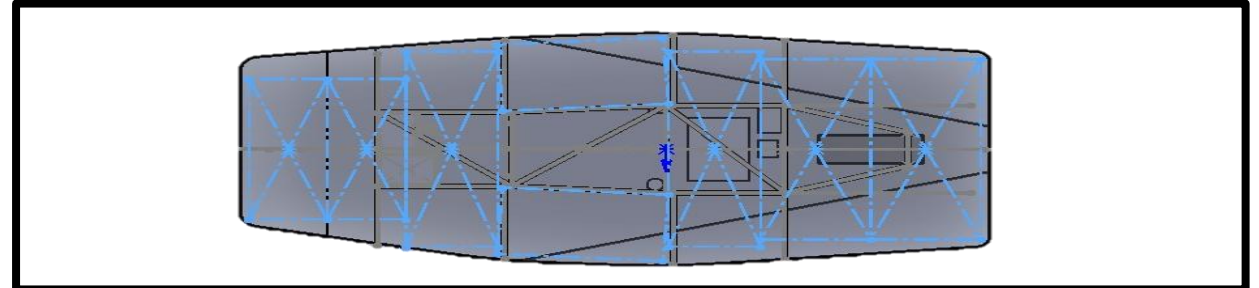
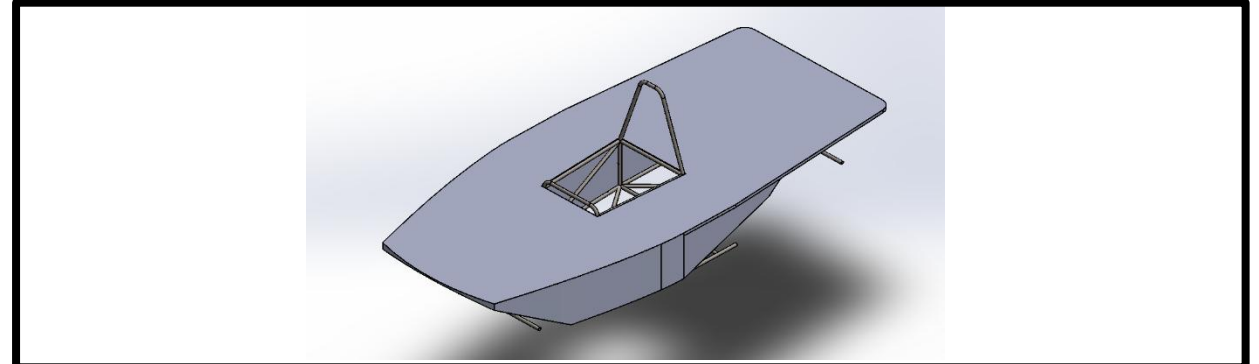
- Type : Monocrystalline
- Number of panels : 8
- At STC power output of the system: 716.88W
- Efficiency : 15.81%

➤ MPPT (Maximum Power Point Tracking) :

- Type : MPPT 100A/20-48V
- Input Voltage : 100V DC (max)
- Output Voltage : 69.6V (max for battery)
- Max Output Current : 20A
- Load Voltage : 48V



- Material used for Bodyworks : Multiwall-Polycarbonate



➤ Panel Configuration

- $V_{oc} = 70.04V$
- $I_{sc} = 12.76A$
- $V_{mp} = 61.8V$
- $I_{mp} = 11.6A$
- $V_{max} = 87.55V$
- $I_{max} = 15.95V$

➤ Solar Cell Specifications

- $V_{oc} = 0.68V$
- $I_{sc} = 6.38A$
- $V_{mp} = 0.6V$
- $I_{mp} = 5.8A$

S.10 Powertrain Calculations & Electrical

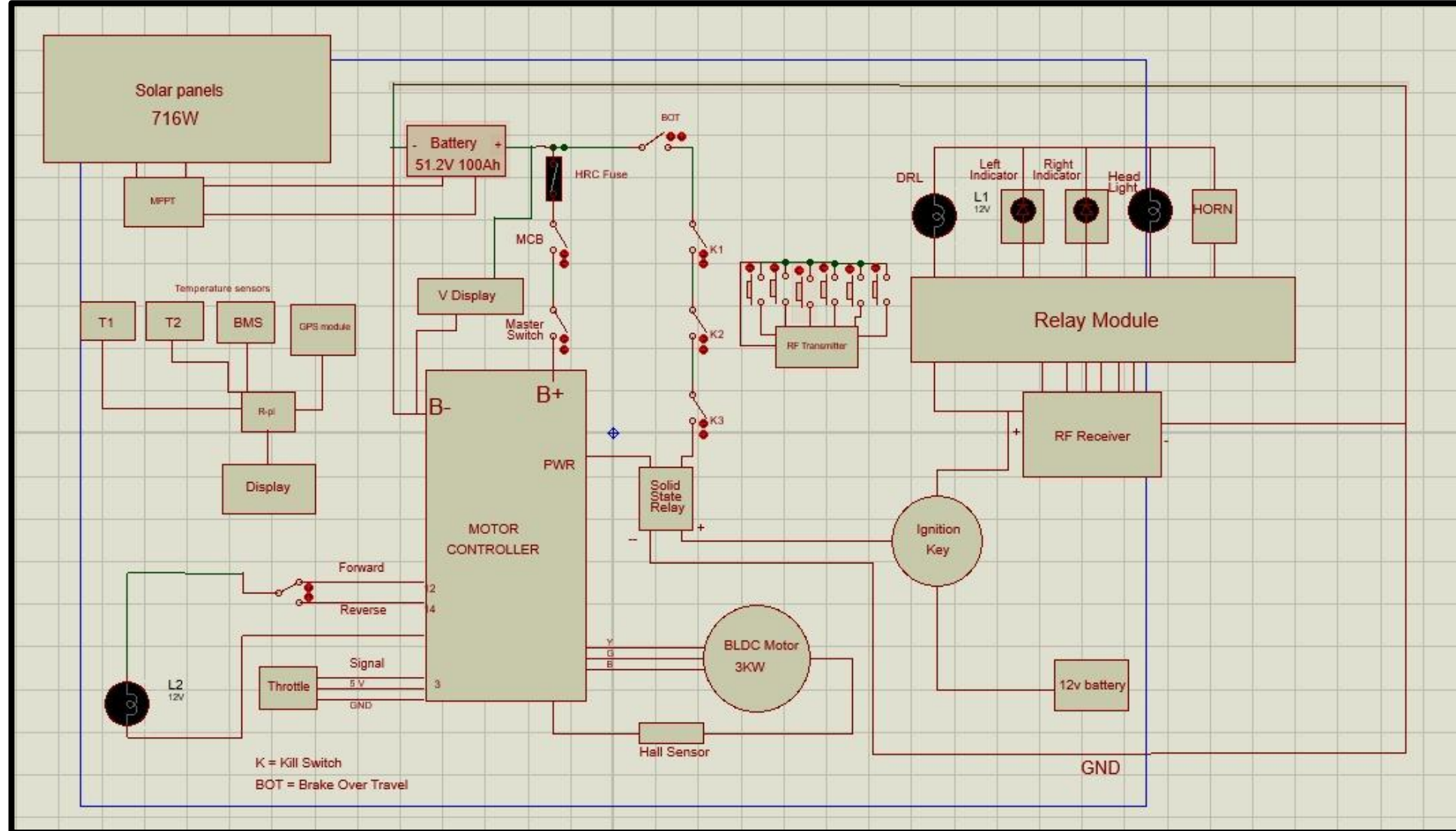
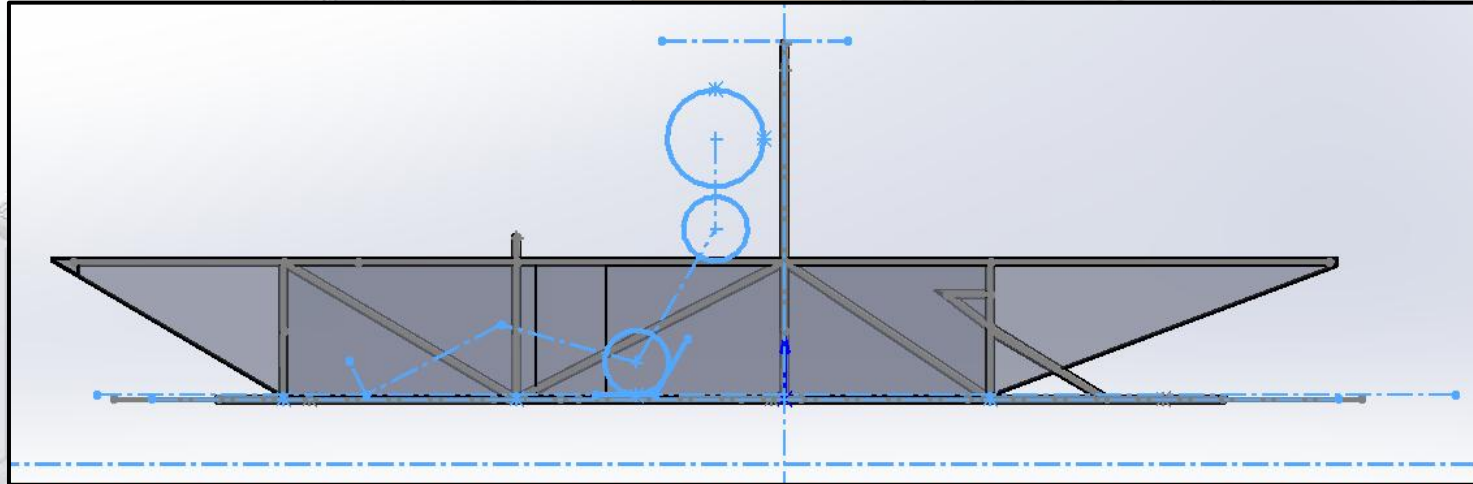


Fig no - 10.1 Tractive System and GLV Block Diagram



S.11 Ergonomics

- **Ergonomics overview :-** Designed for driver comfort, safety & clear visibility, compliant with SEVC rule & SAE 95th percentile male and 5th percentile female standards.



Rider Compartment Safety:

Floor closeout rigidity and thickness were verified through material checks to ensure zero flex and protection from debris.

Ergonomics:

RULA Analysis resulted in a score, indicating that the driver posture is acceptable and does not require immediate corrective action.

Steering System Design:

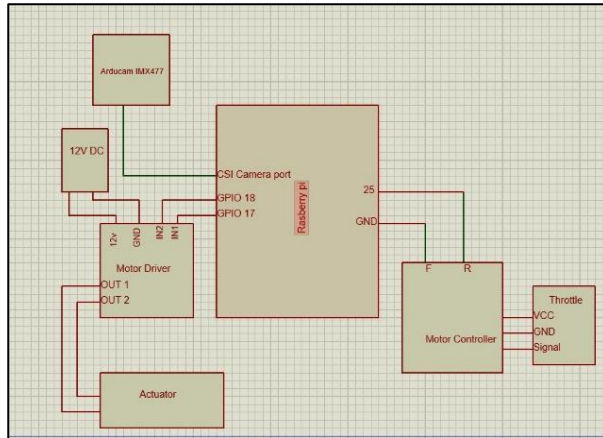
Mechanical rack-and-pinion steering was validated through geometry analysis to ensure precise and reliable control.

Electrical & System Layout:

System schematics and safety checks were used to validate proper integration, isolation, and accessibility of all electrical components.



Autonomous System Architecture



S.12 Autonomous Driving

Alterable (manual to autonomous mode and vice versa):

To changeover from manual to autonomous level, implementation of a toggle switch for easy to access to the driver is implemented.

Level of autonomous driving

Level of automation used: level2

L2 (Level 2) automation includes supervisory control systems like SCADA, which oversee multiple L1 devices and provide an overview of the entire process.

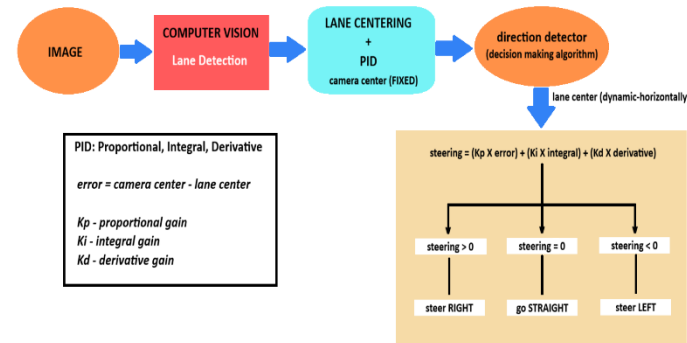
Autonomous Level 2 is widely adopted as it provides steering and speed assistance with driver supervision, offering a practical, cost-effective, and market-ready automation solution.

Autonomous driving Introduction

Self-driving vehicles use sensors, cameras, and AI to navigate without human input.

COMPONENTS	SPECIFICATIONS
Raspberry Pi 4B	Central processing unit for high-level control, data processing, and system monitoring
12 V Relay	Electrical isolation and switching for safety circuit
RF Module	Remote and wireless activation of safety circuit
Arduino UNO	Handles low-level signaling and real-time control tasks
L298N H-Bridge Motor Driver	Drives and controls steering motor direction and speed
Electrical Linear Actuator	Provides mechanical actuation for steering mechanism

Design flow



Integration of automation technology in vehicles

Steering (Rack and Pinion System):

The integration of electric linear actuator in rack and pinion steering system represents a significant advancement in automotive technology.





S.13 Innovation

LiDAR Based Speed Control System

➤ Working :

Use LiDAR to scan the road surface continuously.



Measure road height deviations in real-time before the vehicle reaches them.



Microcontroller processes the data and calculates required speed reduction.

Scope of the Innovation :-

- ✓ Real-time detection of road irregularities.
- ✓ Automatic speed adjustment to enhance safety and comfort.

- Unlike existing mechanical or camera-based methods, LiDAR is non-contact, real-time, and accurate.

➤ Perks...

Ensures passenger comfort.

Reduces suspension and vehicle damage.

Enhances road safety and reduces accident risks.

Adds intelligent automation for modern vehicle systems.

Can be integrated with autonomous driving modules.

Implementation of Touch Sensor Based Intelligent Speed Control System

➤ Working :

Touch sensor on the steering wheel to detect hand placement.



Automatic reduction of vehicle power if the alert is ignored.



A control unit that processes sensor input and triggers safety interventions automatically.

Scope of the Innovation :-

- ✓ Enhances driver safety by ensuring hands remain on the steering wheel.
- ✓ Provides automatic intervention in case of negligence

- Can be implemented in any vehicle, including student-built or commercial vehicles.
- Fully automatic intervention enhances safety without relying on driver reaction.

➤ Perks :

Improves driver safety and reduces accident risk.

Encourages responsible driving behavior.

Provides a low-cost solution for both personal and fleet vehicles.

Can be easily retrofitted in existing vehicles.



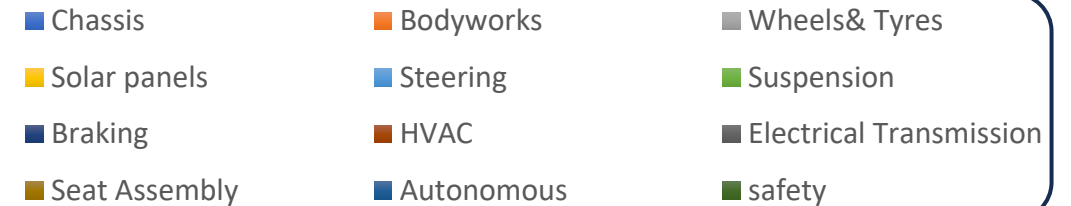
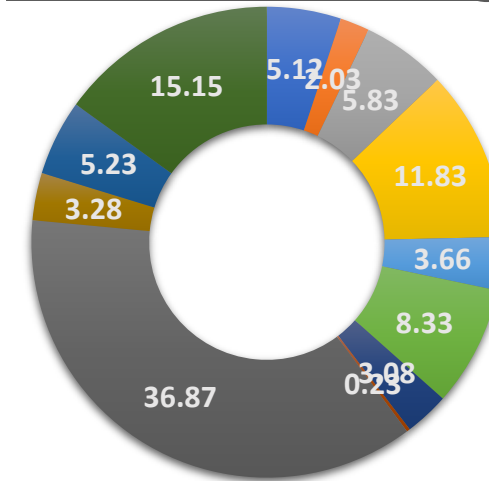
S.14 BUDGET ESTIMATION

SUB-SYSTEM	COST
CHASSIS	30743.00
BODY WORKS	12200.00
WHEELS & TYRES	30500.00
SOLAR PANELS	71000.00
STEERING	22000.00
SUSPENSION	50000.00
BRAKING	18506.00
HVAC	1700.00
ELECTRICAL TRANSMISSION	221217.00
SEAT ASSEMBLY	19706.00
AUTONOMOUS	31450.00
SAFETY	90900.00

**TOTAL
COST**

599922.00

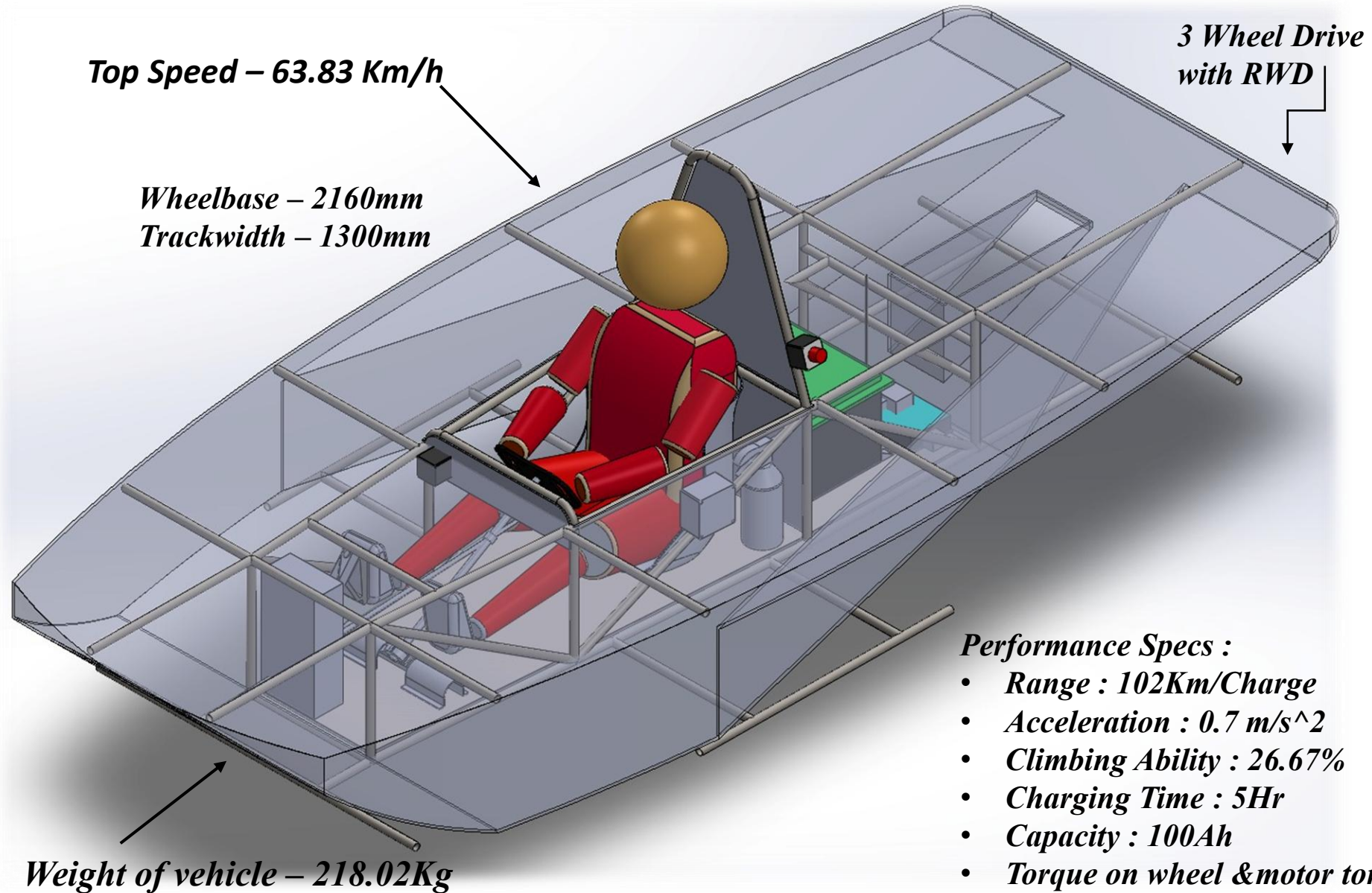
Cost Percentage



SUB-SYSTEM	PRODUCTION COST
CHASSIS	5570.00
BODY WORKS	870.00
TOTAL	6440.00



S.15 Technical and Performance specifications



S.16 Component Images



Fig No. 16.1 Actuator



Fig No. 16.2 Knuckle



Fig No. 16.3 Master Cylinder



Fig No. 16.4 Battery



Fig No. 16.5 Hub Motor



Fig No. 16.6 Motor Controller



Fig No. 16.7 Steering